

CMP-14 - Action and Transaction Engine

Coordinate authorized Actions and Transactions across prepare, commit, observation, rollback, and compensation boundaries.

Authority and state ownership

Book I: Decisions 35, 38-43, 47-51, and 58. Book II: Chapters 16. Authoritative state: Action Records, Transaction Records, Commit Barrier decisions, idempotency records, rollback and compensation state. Explicitly non-owned: intent, authority grants, external-domain policy, outcome evaluation.

Critical invariant

No real effect crosses the Commit Barrier without current authority, verification, exact state, evidence, and Reality Boundary approval.

Logical interfaces

CMP-14-IF-01 / Command / Propose Action

Command `Propose Action` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-02 / Command / Prepare Transaction

Command `Prepare Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-03 / Command / Commit Transaction

Command `Commit Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-04 / Command / Rollback Transaction

Command `Rollback Transaction` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-05 / Command / Begin Compensation

Command `Begin Compensation` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-06 / Query / Get Transaction State

Query `Get Transaction State` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-07 / Query / Explain Commit Decision

Query `Explain Commit Decision` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-08 / Query / Get Idempotency Result

Query `Get Idempotency Result` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-09 / Event / Action Authorized

Event `Action Authorized` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-10 / Event / Transaction Committed

Event `Transaction Committed` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-11 / Event / Transaction Rolled Back

Event `Transaction Rolled Back` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

CMP-14-IF-12 / Event / Compensation Started

Event `Compensation Started` MUST carry identity, authority context where applicable, schema version, correlation, causation, time/freshness, provenance, and explicit success or denial semantics. Book IX owns wire details.

Lifecycle, controls, and recovery

Lifecycle: Proposed > Authorized > Prepared > Committed > Completed; failure to Rolled Back or Compensating. All transitions require version, guard, identity, authority where applicable, idempotency, time context, and evidence. Contain: duplicate effect, partial commit, stale Permission decision, rollback lie, uncontained external effect. Recovery proves identity, state, authority, provenance, dependencies, and invariants before mutation resumes.

Conformance tests

CMP-14-TST-001 - Sole-owner mutation. Attempt mutation through a consumer, replica, cache, operator path, and direct datastore route. Expected: Every unauthorized path is denied and evidenced.

CMP-14-TST-002 - Authority denial. Invoke every command with absent, expired, revoked, mismatched, and over-broad authority context. Expected: No state changes; denial reason and evidence are complete.

CMP-14-TST-003 - Valid lifecycle. Exercise every declared transition with valid guards and fresh versions. Expected: One transition Evidence Object is admitted per transition identifier; duplicate attempts produce no additional state mutation or admitted Evidence Object.

CMP-14-TST-004 - Invalid lifecycle. Attempt skipped, stale, replayed, and terminal-state transitions. Expected: Every invalid transition is rejected without partial mutation.

CMP-14-TST-005 - Critical invariant. No real effect crosses the Commit Barrier without current authority, verification, exact state, evidence, and Reality Boundary approval. Expected: Invariant holds under normal, concurrent, degraded, and recovery conditions.

CMP-14-TST-006 - Failure containment. Inject duplicate effect, partial commit, stale Permission decision, rollback lie, uncontained external effect. Expected: Unsafe work stops, evidence is preserved, and declared containment state is observable.

CMP-14-TST-007 - Recovery. Restore from a verified checkpoint, reconcile dependencies, and resume after validation. Expected: Identity, state, authority, provenance, and evidence continuity are proven.

CMP-14-TST-008 - Contract compatibility. Run current and prior supported contract versions plus an unsupported version. Expected: Supported behavior remains compatible; unsupported input fails explicitly.

CMP-14-TST-009 - Privacy and security. Inject secrets, personal data, malicious input, confused-deputy requests, and log-exfiltration probes. Expected: Least privilege and data controls hold; prohibited data is absent from telemetry.

CMP-14-TST-010 - Topology independence. Repeat critical tests under permitted single-node, replicated, partitioned, and restored topologies. Expected: Topology does not multiply authority or mutation ownership.

Traceability and certification

Twelve requirements map to Books I-III and Book X. No open finding or Owner Review item. Final for Book IV v1.0.